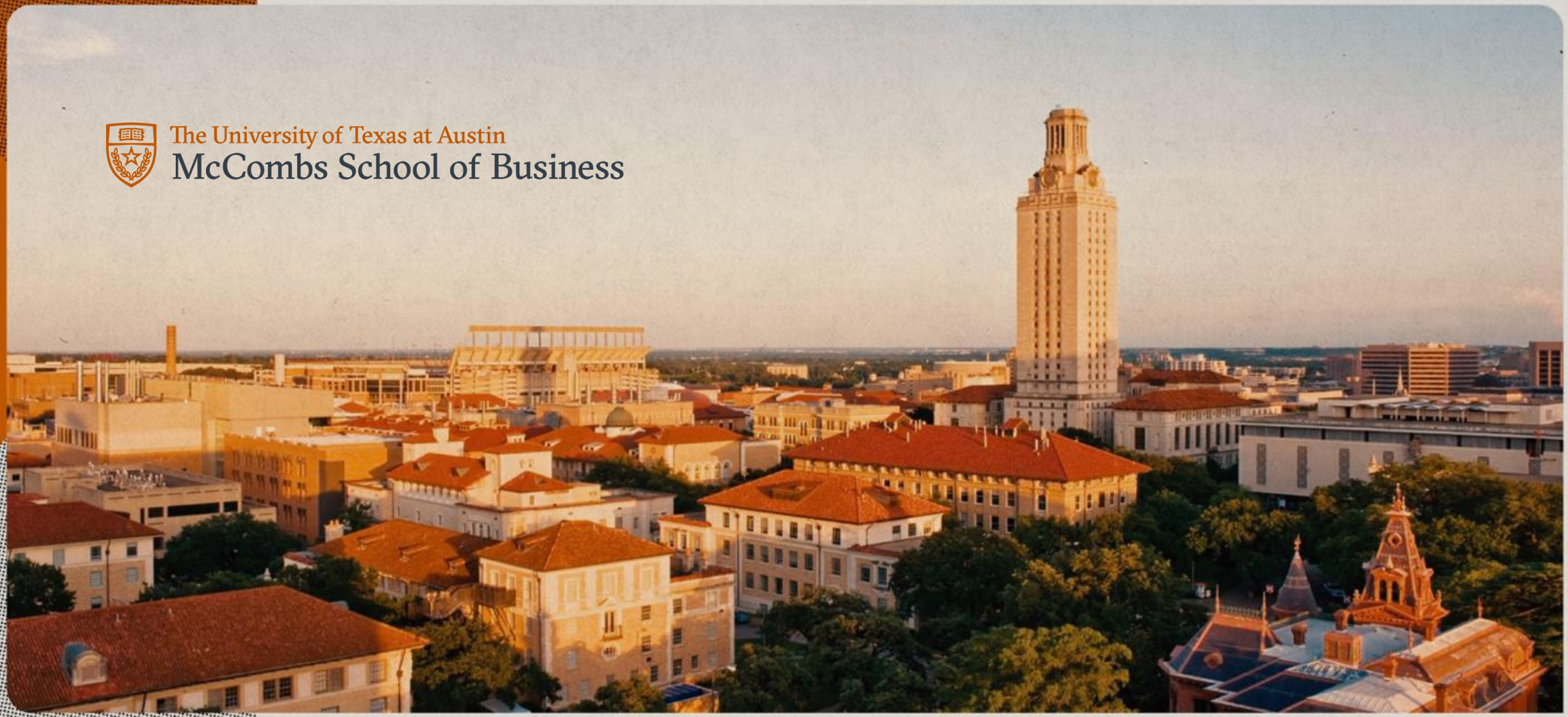


D11: Outliers – Isolation Forest

Outliers from 40,000 feet

In this discussion, I address 4 questions:

1) What are outliers?
2) What problems do outliers create?
3) How can we detect outliers
4) What can we do about outliers?

In general, an **outlier** is a case/record/observation that is not typical of the rest of the data. But there are subtle differences in the meaning we ascribe to “outlier”. An **outlier** could mean only that the data value is “far” from average, and not necessarily that there is anything wrong with it otherwise. Then outliers will occur naturally in proportion to the number of data.

Meaning of Outlier

Meaning 1: Absolute standard

An outlier is a value that exceeds a fixed level – often in relation to a mean value, like 2 or more stdevs from the mean.

Then the number of outliers grows with the size of the data.

Meaning 2: Relative standard

An outlier is a value that exceeds a level expected for the size of the data

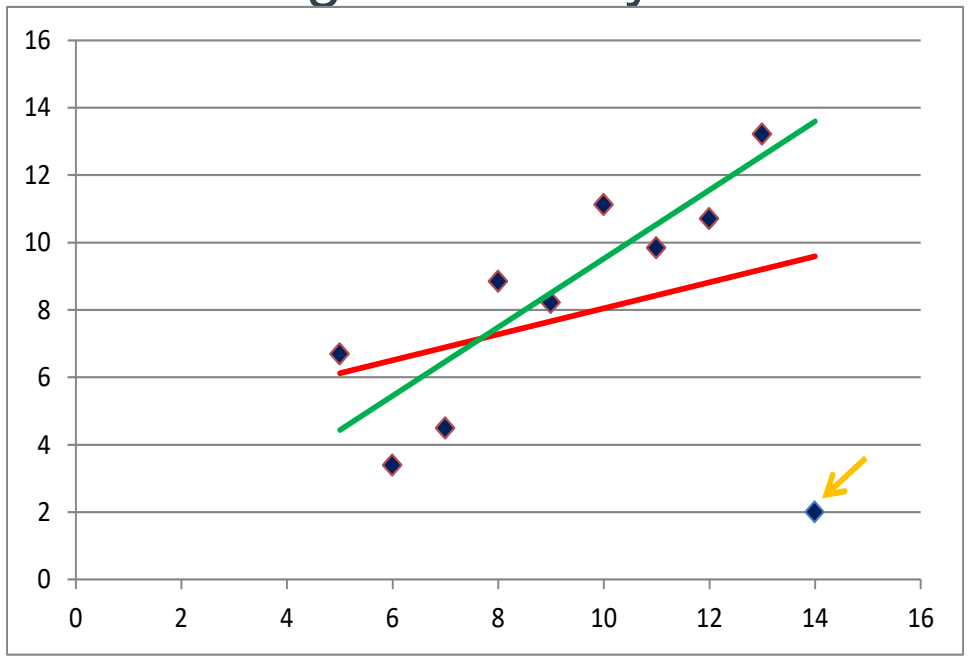
Meaning 3: Objective standard

An outlier is a value that is known not to belong for an outside objective reason, like a house being in Dallas when your study is of Austin

Meaning of Outlier

Meaning 4: Effect

An outlier is a value that would change the analysis substantially if omitted from the data – or would change the analysis substantially if added to data that exclude it.



The regression line is the red line with the datum at the yellow arrow; the regression line is the green line without the datum at the yellow arrow.

Outliers and Influence

Influence

There is a better term for meaning 4 than “outlier”. That is, **influence**. An **influential observation** is a datum that would change the analysis substantially if omitted from the data – or would change the analysis substantially if added to data that exclude it.

An outlier is not necessarily influential.

An influential datum is not necessarily an outlier.

An outlier that is not influential is not a problem.

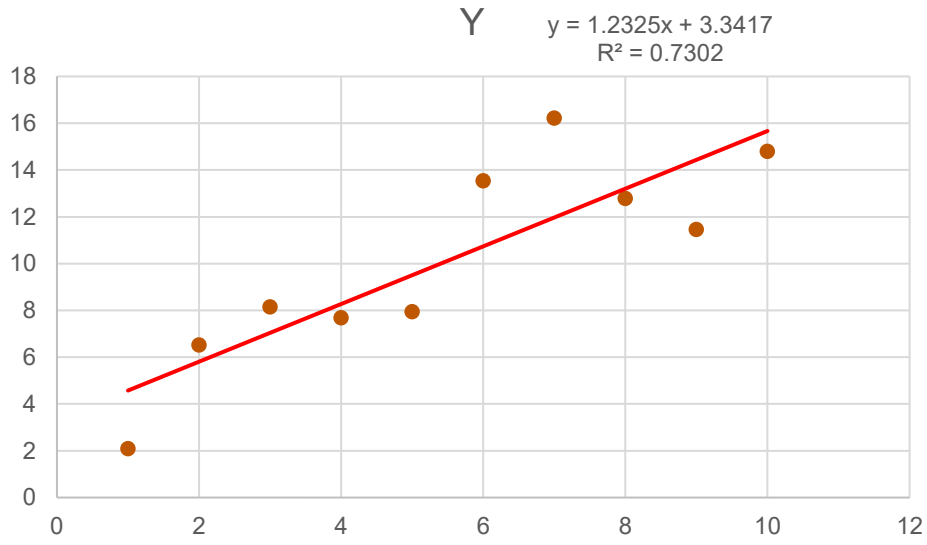
An outlier that is influential is a potential problem.

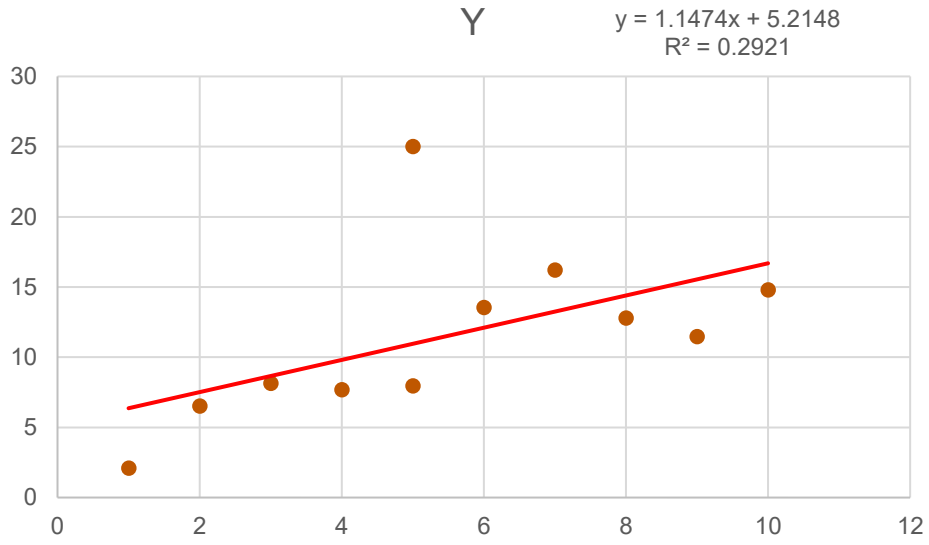
Outliers and Influence

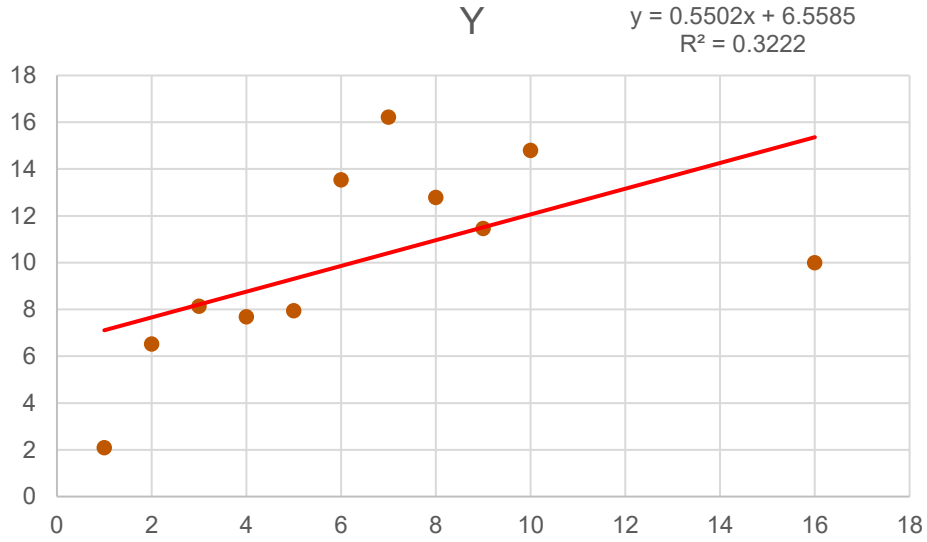
Influence is not bad in itself. In fact, the influential data are the backbone of every analysis – the non-influential data just take up space. So you want your influential data to be “good” data.

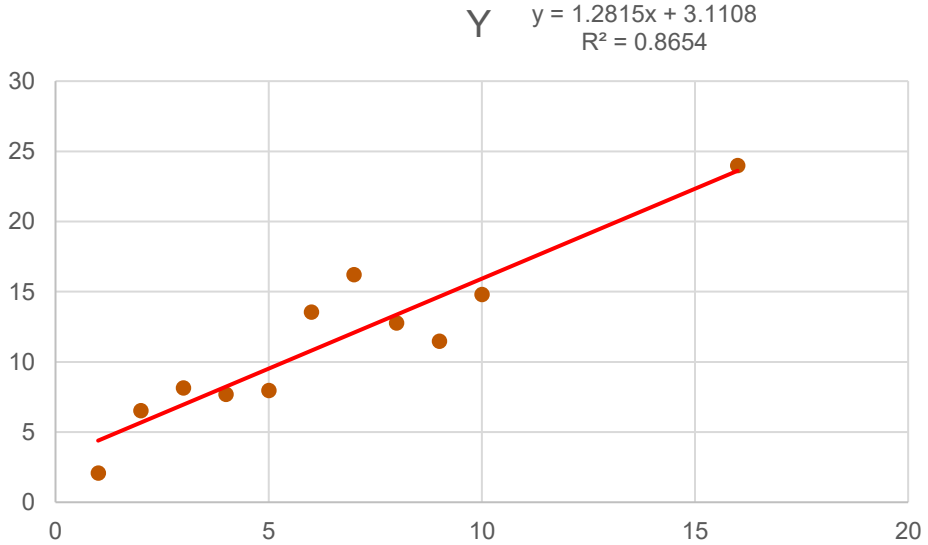
In regression, an observation can be an outlier in X or an outlier in Y or in both (or neither).

Outliers and Influence









What to do about Outliers

The problem with outliers is not that they deviate, but that they may be influential – and not even that they may be influential, but that they may be inappropriately influential.

Drop them – a radical but common remedy. Presumptuous – says your model is right and data are wrong.

Drop them if objectively justified.

Keep them, but turn them into predictors (for regression) – 0/1 indicator variables. This effectively makes each a special cause.

Basic Ideas of Isolation Forest

The working assumption of IF is that anomalies are both rare and different from the other data.

IF uses a binary tree structure to partition the data recursively until each data point is in its own isolated singleton cluster.

The key idea is that anomalies require few partitions to isolate them.

So the count of the number of partitionings needed to isolate each point can be used to classify each data point into few partitionings (outliers/anomalies) and more partitionings (normal data).

The Isolation Forest Algorithm

Randomly select a feature (variable), then randomly select a split value between the min and max of the selected feature. Partition the dataset at the split value into two clusters. Repeat until each data point is in a singleton cluster by itself. The result is an isolation tree. This is a form of hierarchical divisive cluster analysis.

IF produces an ensemble of such isolation trees.

The anomaly score for a data point is based on the path length from the root (whole data set) to the leaf (singleton cluster of the isolated data point), averaged across the ensemble of isolation trees. The shorter the path, the more likely the datum is an anomaly/outlier. <*See "Isolation Forest example with Stock data.xlsx"*>

The Anomaly Score

Mathematically, the anomaly score $s(x, n)$ for a data point x in a dataset of size n is given by:

$$s(x, n) = 2^{-\frac{E(h(x))}{c(n)}}$$

where $E(h(x))$ is the average path length of x across all trees, and $c(n)$ is the average path length of unsuccessful searches in a Binary Search Tree, which can be approximated as:

$$c(n) = 2H(n-1) - \frac{2(n-1)}{n}$$

with $H(i)$ being the harmonic number.

The harmonic number $H(i)$ is defined as the sum of the reciprocals of the first i natural numbers. Mathematically, it is expressed as:

$$H(i) = \sum_{k=1}^{i} \frac{1}{k}$$

Extended Isolation Forest

The "`isotree`" package in R offers an improvement (extension) of the Isolation Forest algorithm. Whereas the original IF algorithm splits the data at hyperplanes that are perpendicular to the split feature, EIF splits by more general hyperplanes. A hyperplane is defined by selecting a random subset of features and a random split value for each feature. Otherwise, the algorithms are the same. General hyperplanes allow for more flexible and effective partitioning of the data space, especially in high-dimensional datasets.

Comparing EIF and IF

EIF Advantages

- More effective identification of anomalies in high dimensional data.
- More flexible partitioning of the data space.

EIF Disadvantages

- Increased complexity, adding to time and resource requirements
- Parameter sensitivity such as the number of features used for hyperplane splits

Major Uses for Isolation Forests

- Fraud detection: Fraudulent transactions in financial systems.
- Network security: Unusual patterns in network traffic that may indicate security breaches.
- Manufacturing: Monitoring machinery for early signs of failure
- Healthcare: Early diagnosis based on anomalies in patient data.

Isolation Forest in R

R implements IF and EIF in the package "`isotree`", available from CRAN, in the `isolation.forest()` function. Its key parameters are:

1. **`ntrees`**: Number of trees in the forest. Default is 100.
2. **`sample_size`**: Number of samples to draw from the dataset to train each tree. Default is "auto", which sets it to the minimum of 256 or the number of samples.
3. **`ndim`**: Number of features to use for each split. Setting this to 1 uses the original Isolation Forest, while values greater than 1 use the Extended Isolation Forest.
4. **`max_depth`**: Maximum depth of the trees. Default is "auto", which allows trees to grow until all points are isolated.
5. **`nthreads`**: Number of threads to use for parallel processing. Default is 1.
6. **`missing_action`**: How to handle missing values. Options include "fail", "impute", and "divide".